

	NOTA TECNICA TECHNICAL REPORT		Núm. No.	ME-T-NT-140060
	Departamento Department	Stress Methods (MESAT)	Avión Aircraft	Technologies
<u>Título/Title</u> Program CRIPCO User's Manual. Version 3				
<u>Palabras clave/Key words</u> SOFTWARE, COMPOSITE, BUCKLING, ALLOWABLES, CRIPPLING			Clasificación acceso Access class P1	
			Registro de Revisiones Revision Record Pag. 2	
<u>Resumen/Summary</u> The purpose of this technical report is to define the way to use the CRIPCO program, version 3. This program calculates the crippling allowable strain for stiffeners made of composite material. Considered stiffeners may be either riveted or cobonded or cocured to skin. Input data and output results are given be means of text files, and a graphics user interface.				
PROPIEDAD DE EADS CASA, Sociedad Unipersonal. Este documento no puede ser utilizado ni reproducido total o parcialmente sin la previa autorización escrita de la Dirección de Ingeniería y Tecnología de EADS CASA. EADS CASA, Sociedad Unipersonal PROPERTY. This document shall neither be used nor completely or partially reproduced without previous written authorization by EADS CASA Engineering and Technology Directorate.				
Distribución/Distribution	Realización/Execution		Edición/Issue Fecha edición/Issue date A 14/05/2014	
Structural Analysis (MESA)	Realizado Prepared		Firma Signature	
Airframe Development (COEAD)	José Angel Velasco Vaquero			
	Gustavo Baños Lapuente			
	Comprobado Checked	Gustavo Baños Lapuente		
	Aprobado Approved	Darío Fernández Villalba		

REGISTRO DE REVISIONES/REVISIONS RECORD

[illegible]

INDEX

1	Technical sheet of program	4
2	Introduction	5
3	Execution of the program.....	6
4	Input data file.....	9
5	Output results.....	12
	5.1 Output result file	12
	5.2 Errors:.....	13
6	Windows application.....	7
7	Examples	15
	7.1 Example 1	15
	7.2 Example 2	19
	7.3 Example 3	24
	7.4 Example 4	29
	7.5 Example 5	34
	7.6 Example 6	39
	7.7 Example 7	44
	7.8 Example 8	49
	7.9 Example 9	54

1 Technical sheet of program

Technical Note **ME-T-NT-140060**

<u>PROGRAM</u>	<u>REFERENCE</u>	<u>DATE</u>
CRIPCO		2014

OBJECTIVE

This program calculates the crippling allowable strain for stiffeners made of composite material.

<u>LANGUAGE</u>	<u>COMPUTER</u>
Fortran 77	Sun Solaris
Visual Basic Net	PC

<u>NUMBER OF LINES (APROX.)</u>	<u>EXECUTION MODE</u>
1400 (Fortran Analysis core)	Interactive
1400 (CV.B. Net)	Interactive GUI
2300 (Blank lines and comments excluded)	

<u>AUTHOR</u>	<u>VERSION</u>
José Carlos Gómez López	1.0
J. A. Velasco / D. Fernández	2.0
José Angel Velasco Vaquero	3.0

ADDITIONAL DOCUMENTS

Technical Note: NT-T-ADS-93060 Ed. B.
"PROCEDIMIENTO DE CALCULO DE CRIPPLING EN MATERIAL
COMPUESTOS"

2 Introduction

This technical report describes the usage procedure for version 3.0 of CRIPCO program. Nine examples are included.

This program calculates the crippling allowable strains for composite material stiffeners. Either riveted or cobonded or cocured stiffeners are considered.

Once the geometry and material data are introduced, the program calculates the local buckling strain and the crippling strain of each segment of the stiffener. This crippling strain is limited to the laminate allowable compression strain.

Only two possible boundary conditions are considered for each segment: free-supported or supported-supported. Other conditions taking into account the stiffness supplied by the corners of the section to each segment is not considered.

The output results for this program are the local buckling microstrains and the crippling microstrains for each segment of the stiffener, as well as the stiffener crippling microstrains.

Differences between versions:

- **Version 1.0:**

Initial version developed in Fortran 77.

Input data are given interactively and the results are shown on the screen.

- **Version 2.0:**

The data flow of the program is modified. Now, the input data as well as the output results are provided by means of files:

- Input file: With the input data grouped in the same way as for other applications.
- Output file: With all the input data and output results in printable form.

The user can choose the language: Spanish or English.

The program is restructured to allow the creation of a dynamic link library DLL for Windows operative system.

In parallel with this, a Windows application has been developed as Graphic User Interface GUI for the program CRIPCO. This application uses the same input file and provides the same output results. The crippling analysis is performed by calling to the DLL, compiled from the same source as the official Fortran version.

- **Version 3.0:**

If the segment with lower crippling allowable is normal to the axis with minimum bending inertia, this minimum strength will be the cut-off value for the rest of the segments in the stringer. If the segment with minimum crippling allowable is parallel to the minimum inertia axis the second minimum crippling strength will be set as cut-off of value for the rest of segments. This is the criterion proposed in NT-T-ADS-93060-B.

The allowable crippling microstrains for the stiffener are obtained as the average of the allowable of all the segments that form the stiffener, taking into account the modulus and area of the segments.

3 Execution of the program

3.1 Multivac Computer Version

Prior to the program execution, it is necessary to create the input data file, as described in section 4. The program can be then executed by typing its full path name.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

```
$scripco
```

The program asks for the language to use, the name of the input file and the name of the two output files.

```
Idioma / Language:
=====
      1 ... ESPAÑOL
      2 ... ENGLISH
Idioma / Language: 2
Name of the input file: Example.dat
Name of the output file: Example.txt
```

Details of the input and output files are provided in next sections. The CPU time for calculation is negligible.

It is possible to execute the program with a single command, including the input and output filenames and the language as parameters. The above example can be executed as follows:

```
$scripco Example.dat Example.txt
```

If the third parameter (language) is omitted, the Spanish language will be taken by the program.

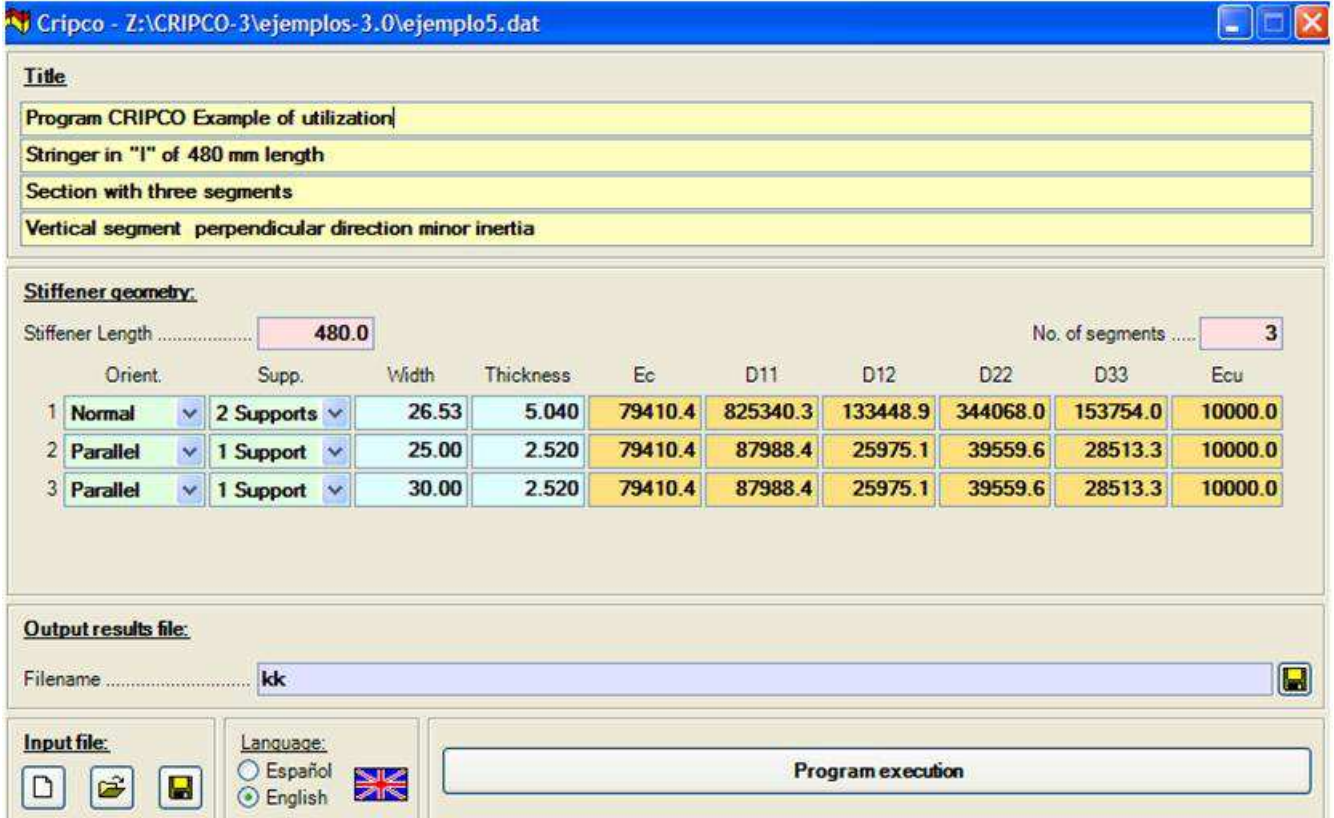
4 Windows application

An application has been developed in Visual Basic for Windows environment. The original routines of the program for input data, calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please consult with the system manager,

The application can be executed by "double click" in the file **Cripco.exe**, located in the PC's server, or by means of a **link** in the "Desktop" or in the "Start Menu".

The input data can be retrieved from an input data file or directly typed in the application form. The form has the following appearance:



Cripco - Z:\CRIPCO-3\ejemplos-3.0\ejemplo5.dat

Title

Program CRIPCO Example of utilization

Stringer in "I" of 480 mm length

Section with three segments

Vertical segment perpendicular direction minor inertia

Stiffener geometry:

Stiffener Length **480.0** No. of segments **3**

	Orient.	Supp.	Width	Thickness	Ec	D11	D12	D22	D33	Ecu
1	Normal	2 Supports	26.53	5.040	79410.4	825340.3	133448.9	344068.0	153754.0	10000.0
2	Parallel	1 Support	25.00	2.520	79410.4	87988.4	25975.1	39559.6	28513.3	10000.0
3	Parallel	1 Support	30.00	2.520	79410.4	87988.4	25975.1	39559.6	28513.3	10000.0

Output results file:

Filename **kk**

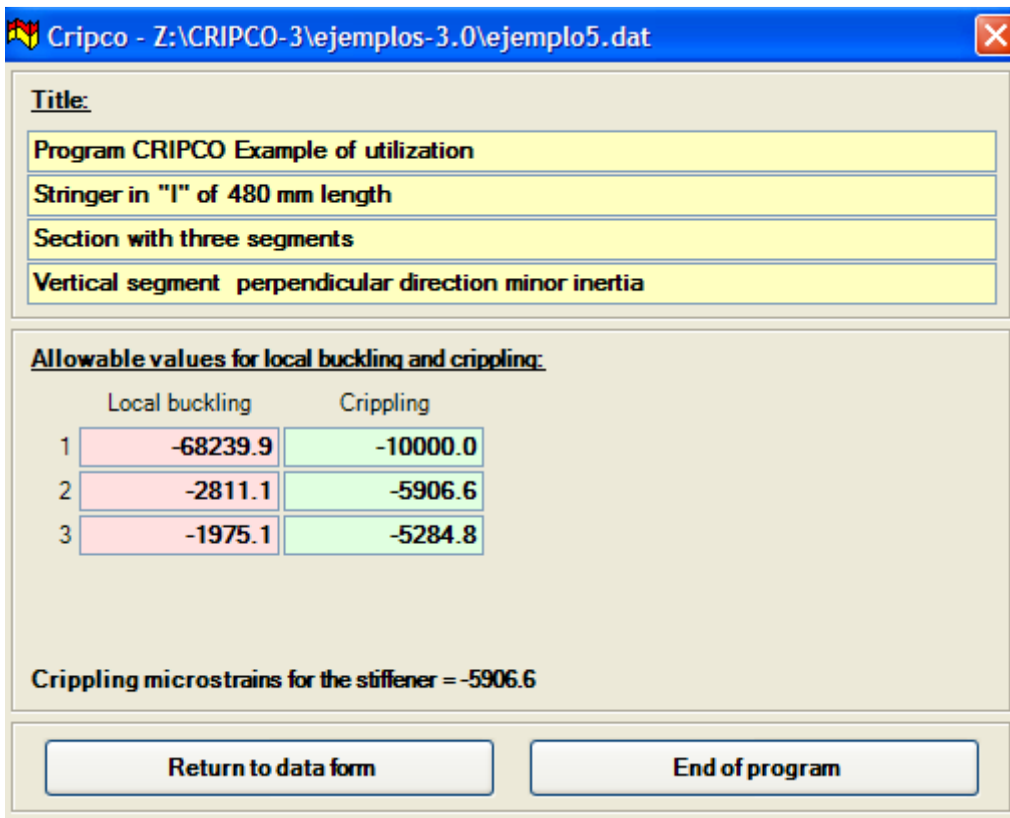
Input file:

Language: ☐ Español ☒ English

Program execution

Figure 1. Input data form.

The program can be executed clicking the button "Program execution". Then the application solves the problem, writes the output files and shows the results in the following form:



CRIPCO - Z:\CRIPCO-3\ejemplos-3.0\ejemplo5.dat

Title:

Program CRIPCO Example of utilization

Stringer in "I" of 480 mm length

Section with three segments

Vertical segment perpendicular direction minor inertia

Allowable values for local buckling and crippling:

	Local buckling	Crippling
1	-68239.9	-10000.0
2	-2811.1	-5906.6
3	-1975.1	-5284.8

Crippling microstrains for the stiffener = -5906.6

Return to data form **End of program**

Figure 2. Output results form.

The configuration considered for this example is shown in next figure:

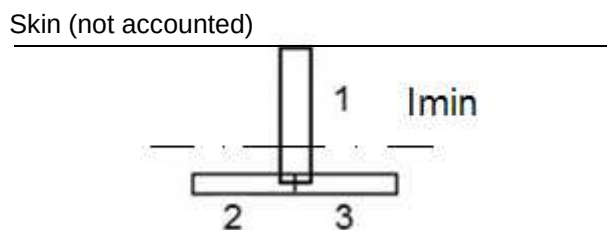


Figure 3. Geometry for windows application example

5 Input data file

This file contains all the necessary data that define the problem, as well as the different options required for solving it.

Basically, it contains four data groups, all of them are necessary. They will be numbered from 1 to 4 and must be in order.

All the numerical data contained in this file will be read in a free format. The variables will be separated by one or more blank spaces. The numbering of each group will be included in the first column.

Listed below are the contents of each data group:

GROUP 1: Title of the problem:

Four lines containing the title of the problem to be solved.

Line 1 -----> 1 First line of the title
Line 2 -----> Second line of the title
Line 3 -----> Third line of the title
Line 4 -----> Fourth line of the title

GROUP 2: Length of the stiffener:

This group has three lines with the following information

Line 1 -----> 2 "Optional comment"
Line 2 -----> Length

Where Length = Spacing between supports for the stiffener (e.g.: ribs, frames)

GROUP 3: Number of segments:

The structure of this group is the following:

Line 1 -----> 3 "Optional comment"
Line 2 -----> Ns

Where Ns = Number of segments in which the stiffener can be divided.

Several considerations must be taken into account for the segments adjacent to the panel.

- In the case the panel thickness is different at each side of the stiffener, the thickest panel will be extended until it completely covers the stiffener segment.

- If the stiffener is riveted to the panel, the segment is considered separately (see Figure 4.).

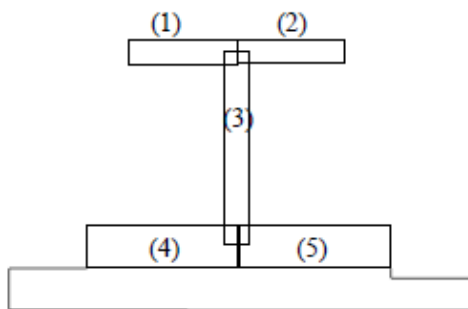


Figure 4. Stiffener riveted to the skin

- If the stiffener is cobonded or cocured to the skin, two possible configurations may take place:
 - If the overall thickness is equal or greater than twice the maximum skin thickness, $t_t \geq 2 \cdot \max(t_{sk1}, t_{sk2})$, the skin is assumed to belong to the stiffener, making up a unique segment with it (see Figure 5. right side).
 - On the other hand, if the overall thickness is lower than twice the maximum skin thickness, $t_t < 2 \cdot \max(t_{sk1}, t_{sk2})$, the stiffener segment is assumed to belong to the skin, so no segment is considered for crippling or local buckling calculation (see Figure 5. left side).

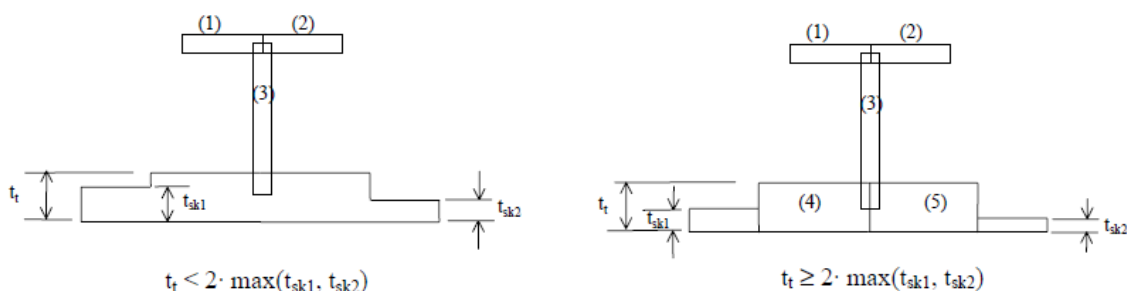


Figure 5. Stiffener cocured or cobonded to skin

GROUP 4: Segment data

This group has N_s+1 lines with the following content:

Line 1 -----> 4 "Optional commentary"

Line 2 -----> $I_p(1)$ $I_s(1)$ $W(1)$ $t(1)$ $E_c(1)$ $D_{11}(1)$ $D_{12}(1)$ $D_{22}(1)$ $D_{33}(1)$ $\epsilon_c(1)$

Line 3 -----> $I_p(2)$ $I_s(2)$ $W(1)$ $t(2)$ $E_c(2)$ $D_{11}(2)$ $D_{12}(2)$ $D_{22}(2)$ $D_{33}(2)$ $\epsilon_c(2)$

.....

Line N_s-1 ---> $I_p(N_s)$ $I_s(N_s)$ $W(N_s)$ $t(N_s)$ $E_c(N_s)$ $D_{11}(N_s)$ $D_{12}(N_s)$ $D_{22}(N_s)$ $D_{33}(N_s)$ $\epsilon_c(N_s)$

Where: $Ip(i)$ = Orientation of the segment "i". It is an integer with the following meaning:

$Ip(i) = 1$ If the segment is parallel to the direction of minor inertia.

$Ip(i) = 2$ If the segment is perpendicular to the direction of minor inertia.

The inertia properties can be obtained by running AIDA program using thickness normalized with Young Modulus.

$Is(i)$ = Support condition for segment "i". It is an integer with the following meaning:

$Is(i) = 1$ If the segment has one side supported and the other free, as segments (1) and (2) in previous figures.

$Is(i) = 2$ If the segment has both sides supported, as segment (3) in previous figures.

$W(i)$ = Is the width of the segment, measured along the middle line of the section.

$t(i)$ = Is the thickness of the segment. For stiffeners cocured or cobonded to the skin, the same considerations made for the number of segments definition must be observed.

$E_c(i)$ = Is the Young modulus ~~under compression~~, corresponding to the mean properties of the laminate.

$D_{xy}(i)$ = Are the terms D_{11} , D_{12} , D_{22} and D_{33} of the flexural stiffness matrix $[D]$, under compression, corresponding to B-values of the laminate.

$\epsilon_c(i)$ = Are the allowable microstrains under compression, corresponding to B-values of the laminate. It is the upper limit for the crippling microstrains of each segment.

• Laminate Young modulus shall be calculated from lamina moduli averaged between tension and compression (mean basis)

• Stiffness matrix D shall be calculated from lamina moduli in compression as B-Values

6 Output results

This section includes the description of the output result file as well as the possible error messages.

6.1 Output result file

This file provides the following information:

- **Summary of the input data:**
 - = Title.
 - = Stiffener Length
 - = Number of segment
 - = Segment data
- **Allowable microstrains for each segment:**
 - = Allowable microstrains for Local Buckling of the segment.
 - = Allowable microstrains for Crippling of the segment. Truncated to the critical one, (first minimum is the critical segment is normal to direction of minimum inertia, second minimum is the critical segment is parallel to direction of minimum inertia)
- **Allowable microstrains for the stiffener:**
 - = Allowable microstrains for Crippling of the stiffener. Obtained as:

$$\epsilon_{all, stiffener} = \frac{\sum_i E c_i \times W_i \times t_i \times \epsilon_{all_i}}{\sum_i E c_i \times W_i \times t_i}$$

For i = 1 to Ns

6.2 Errors:

When the program detects an error in the input data or writing the results, it writes the corresponding message in the console and aborts the execution of the program.

The error messages that might appear in the console are listed below:

6.2.1 Error messages. English output:

ERROR: The input file cannot be opened
ERROR: Erroneous group number in the input file
ERROR: Group 1 missing in input file (Title)
ERROR: End of file while reading group 1 in the input file (Title)
ERROR: Group 2 missing in the input file (Length)
ERROR: Reading error while reading group 2 in the input file (Length)
ERROR: The length must be greater than 0
ERROR: Group 3 missing in input file (No. of segments)
ERROR: Reading error while reading group 3 in the input file (No. of segments)
ERROR: The number of segments must be greater than 0
ERROR: Group 4 missing in input file (Segment data)
ERROR: Reading error while reading group 4 in the input file (Segment data)
ERROR: Error in the segment position. It must be 1 or 2
ERROR: Error in the definition of the support condition. It must be 1 or 2
ERROR: The mid-line width of the segment must be greater than 0
ERROR: The thickness of the segment must be greater than 0
ERROR: The mean value of E_c for the laminate must be greater than 0
ERROR: The allowable value D11 for the laminate must be greater than 0
ERROR: The allowable value D12 for the laminate must be greater than 0
ERROR: The allowable value D22 for the laminate must be greater than 0
ERROR: The allowable value D33 for the laminate must be greater than 0
ERROR: The allowable microstrains in compression E_{cu} for the laminate must be greater than 0
ERROR: The output file cannot be opened
ERROR: Writing error in the output file

6.2.2 Error messages. Spanish output:

ERROR: No se puede abrir el fichero de entrada
ERROR: Número de grupo erróneo en el fichero de entrada
ERROR: Grupo 1 ausente en el fichero de entrada (Título)
ERROR: Fin de fichero mientras se leía el grupo 1 en el fichero de entrada (Título)
ERROR: Grupo 2 ausente en el fichero de entrada (Longitud)
ERROR: Error de lectura mientras se leía el grupo 2 en el fichero de entrada (Longitud)
ERROR: La longitud debe ser mayor que 0
ERROR: Grupo 3 ausente en el fichero de entrada (Nº de tramos)
ERROR: Error de lectura mientras se leía el grupo 3 en el fichero de entrada (Nº de tramos)
ERROR: El número de tramos debe ser mayor que 0
ERROR: Grupo 4 ausente en el fichero de entrada (Datos de tramo)
ERROR: Error de lectura mientras se leía el grupo 4 en el fichero de entrada (Datos de tramo)
ERROR: Error en la posición del tramo. Debe ser 1 ó 2
ERROR: Error en la definición de la condición de apoyo. Debe ser 1 ó 2
ERROR: La anchura media del tramo debe ser mayor que 0
ERROR: El espesor del tramo debe ser mayor que 0
ERROR: El valor medio de Ec para el laminado debe ser mayor que 0
ERROR: El valor admisible D11 para el laminado debe ser mayor que 0
ERROR: El valor admisible D12 para el laminado debe ser mayor que 0
ERROR: El valor admisible D22 para el laminado debe ser mayor que 0
ERROR: El valor admisible D33 para el laminado debe ser mayor que 0
ERROR: Las microdef admisibles a compresión Ecu para el laminado deben ser mayores que 0
ERROR: No se puede abrir el fichero de salida
ERROR: Error de escritura en el fichero de salida

8 Examples

Input data and obtained results are presented for the two following examples.

8.1 Example 1

The case to solve is a T-shape stringer cobbled or cocured to the skin. The geometry is as shown on next figure.

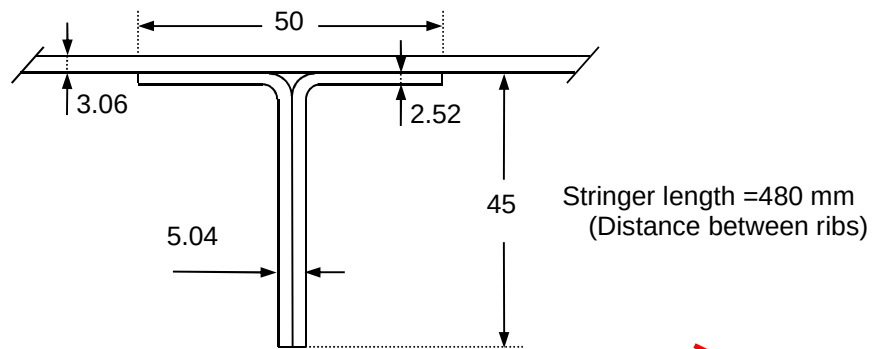


Figure 6. Example 1 geometry.

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ϵ_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

Value used for ply thickness is 0.18mm in this example

- Lay-up: Stringer (+45/-45/0/0/0/90/0)S
Skin (45/-45/0/-45/45/90/-45/+45/0)\$

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = 3.06 \text{ mm}$$

$$t_t = 2.52 + 3.06 = 5.58 \text{ mm}$$

Since $t_t < 2 \cdot t_{sk}$, only the vertical segment is going to be considered.

The dimensions to be considered for this segment are:

$$W = 45 + 3.06 - 5.58/2 = 45,27 \text{ mm}$$

$$t = 5.04 \text{ mm}$$

The lay-up for the vertical segment is: (+45/-45/0/0/0/90/0/0/90/0/0/0/-45/+45)S

The material properties required by the program for the laminate of the considered segment can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

$$\begin{aligned} E_c &= 79410.4 \text{ N/mm}^2 \\ D_{11} &= 825340.3 \text{ N/mm}^2 \\ D_{12} &= 133448.9 \text{ N/mm}^2 \\ D_{22} &= 344068.0 \text{ N/mm}^2 \\ D_{33} &= 153754.0 \text{ N/mm}^2 \\ \epsilon_c &= 10000.0 \text{ } \mu\epsilon \end{aligned}$$

The inertia properties are obtained by running AIDA program. The results are shown in figure below.

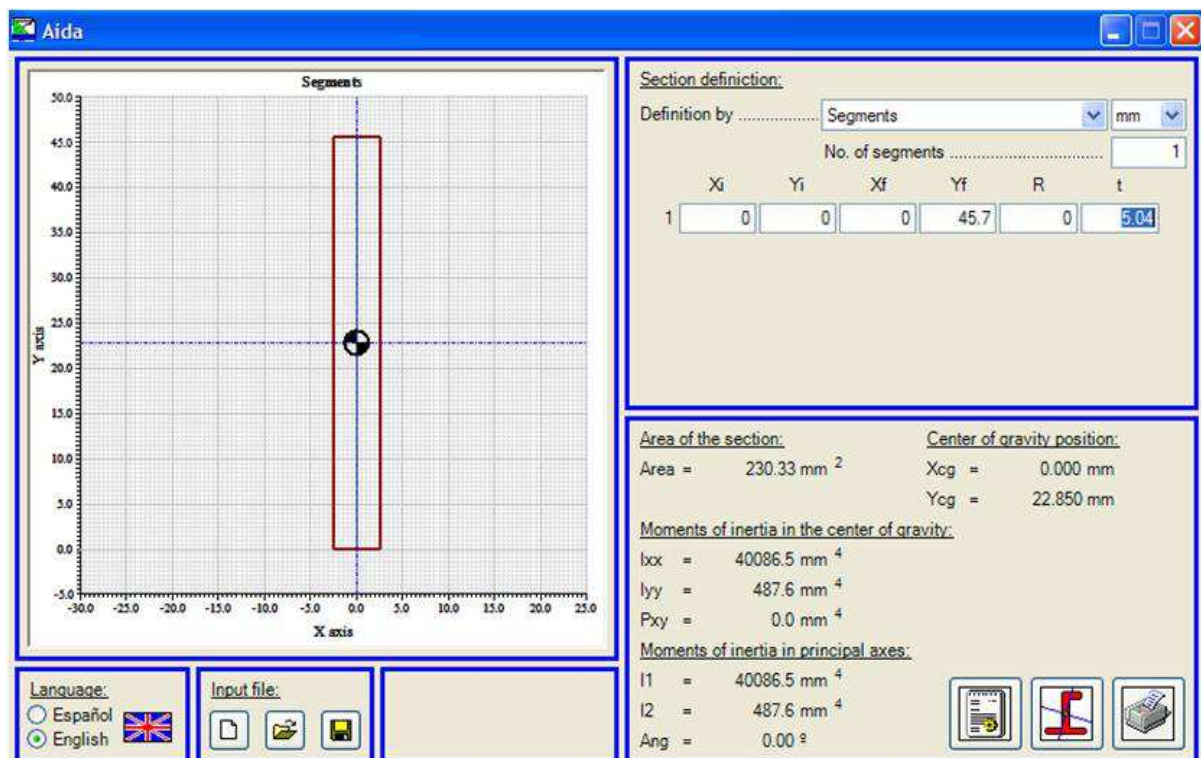


Figure 7. Example 1 Inertia properties

This segment is parallel to the direction of minimum inertia, $I_p = 1$, and it has one end free and the other end supported, $I_s = 1$.

The input and output files corresponding to this example are shown on next pages:

8.1.1 Input file Example 1:

```
1 PROGRAM CRIPCO Example of utilization 1
  Stringer in "T" of 480 mm length
  Section with one segment
  Vertical segment parallel direction minor inertia
2 Longitud
  480.0
3 Número de tramos
  1
4 Datos de tramo: Orient. Apoyo Anchura Espesor Ec D11 D12 D22 D33 Ecu
  1 1 44.73 5.040 79410.4 825340.3 133448.9 344068.0 153754.0 10000.0
```

8.1.2 Output file Example 1

```
*****
*
*          <<< CRIPCO >>>
*
*****
* Calculation of the crippling allowable *
* value for composite stringers        *
```

Title:
=====

PROGRAM CRIPCO Example of utilization 1
 Stringer in "T" of 480 mm length
 Section with one segment
 Vertical segment parallel direction minor inertia

Input data:
=====

Length of the stiffener: 480.0

Number of stiffener segments: 1

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	44.73	5.040	1	1	79410.4	825340.3	133448.9	344068.0	153754.0	10000.0

Allowable values for local buckling and crippling:
=====

Segment	Local buckling	Crippling
1	-2657.5	-5803.0

```
*****
* Crippling microstrains for the stiffener: -5803.0 *
*****
```

8.2 Example 2

The case to solve is a T-shape stringer cobbled or cocured to the skin. The geometry is as shown on next figure.

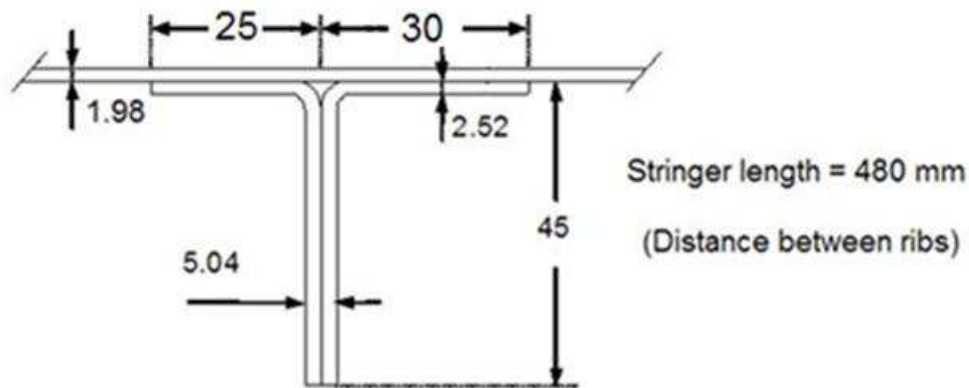


Figure 8. Example 2 Geometry

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ϵ_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

- Lay-up: Stringer (+45/-45/0/0/90/0)S
 Skin (45/-45/0/0/90/0)\$

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = 1.98 \text{ mm}$$

$$t_t = 2.52 + 1.98 = 4.50 \text{ mm}$$

Since $t_t \geq 2 \cdot t_{sk}$, three segments are going to be considered.

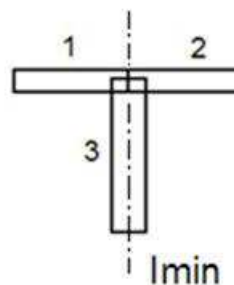


Figure 9. Example 2 : Segments

The dimensions and lay-up to be considered for these segments are:

Segment	W (mm)	t (mm)	Lay-up
1	25	4.5	45/-45/0/-45/45/90/-45/45/0/45/-45/90/45/-45/0/-45/45/ 45/-45/0/0/0/90/0/0/90/0/0/0/-45/45
2	25	4.5	45/-45/0/-45/45/90/-45/45/0/45/-45/90/45/-45/0/-45/45/ 45/-45/0/0/0/90/0/0/90/0/0/0/-45/45
3	$45+1.98-4.5/2=44.73$	5.04	(+45/-45/0/0/0/90/0/0/90/0/0/0/-45/+45)S

The material properties required by the program for the laminate of the considered segments can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

Segment	1 & 2	3
E_c	73937.5 N/mm ²	79410.4 N/mm ²
D_{11}	546605.6 N/mm ²	825340.3 N/mm ²
D_{12}	104087 N/mm ²	133448.9 N/mm ²
D_{22}	267552.7 N/mm ²	344068.0 N/mm ²
D_{33}	118540.7 N/mm ²	153754.0 N/mm ²
ϵ_c	10000.0 $\mu\epsilon$	10000.0 $\mu\epsilon$

The inertia properties are obtained by running AIDA program with thickness normalized with the Young Modulus of segment 1.

Segment	E_c (N/mm ²)	th_{real} (mm)	th_{norm} (mm)
1	73937.5	4.5	4.5
2	73937.5	4.5	4.5
3	79410.4	5.04	5.4

The results are shown in figure below.

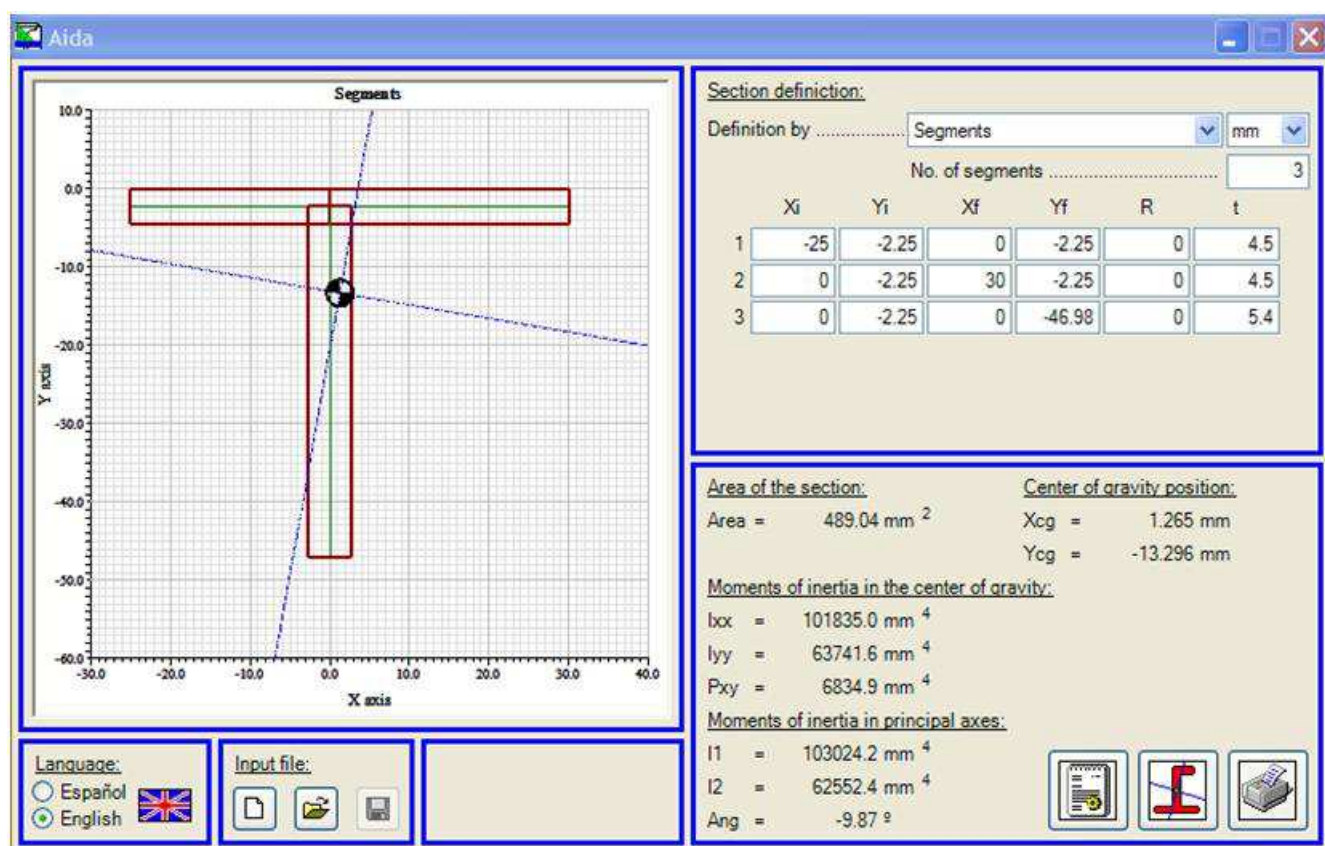


Figure 10. Example 2 Inertia Properties

Therefore, segments 1 and 2 are perpendiculars to the direction of minor inertia, $I_p = 2$. Meanwhile, the vertical segment number 3 is parallel to the direction of minor inertia, $I_p = 1$.

For all the segments it has one end free and the other en supported, $I_s = 1$.

The input and output files corresponding to this example are shown on next pages:

8.2.1 Input file Example 2:

```

1 PROGRAM CRIPCO Example of utilization 2
  Stringer in "T" of 480 mm length
  Section with three segments
  Vertical segment parallel direction minor inertia
2 Length
  480.0
3 umber of segments
  3
4 Segment data: Orient.  Apoyo  Anchura  Espesor  Ec  D11  D12  D22  D33  Ecu
  2 1   25.00   4.500  73937.5   546605.6   104087.0   267552.7   118540.7   10000.0
  2 1   30.00   4.500  73937.5   546605.6   104087.0   267552.7   118540.7   10000.0
  1 1   44.73   5.040  79410.4   825340.3   133448.9   344068.0   153754.0   10000.0

```

8.2.2 Output file Example 2

```

*****
*
*          <<< CRIPCO >>>
*
*****
* Calculation of the crippling allowable
* value for composite stringers
*****

```

Title:
=====

PROGRAM CRIPCO Example of utilization 2
 Stringer in "T" of 480 mm length
 Section with three segments
 Vertical segment parallel direction minor inertia

Input data:
=====

Length of the stiffener: 480.0

Number of stiffener segments: 3

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	25.00	4.500	2	1	73937.5	546605.6	104087.0	267552.7	118540.7	10000.0
2	30.00	4.500	2	1	73937.5	546605.6	104087.0	267552.7	118540.7	10000.0
3	44.73	5.040	1	1	79410.4	825340.3	133448.9	344068.0	153754.0	10000.0

Allowable values for local buckling and crippling:

=====

Segment	Local buckling	Crippling
1	-7122.1	-7096.5
2	-5031.9	-7096.5
3	-2657.5	-5803.0

```

*****
* Crippling microstrains for the stiffener: -6456.8 *
*****

```

8.3 Example 3

The case to solve is a T-shape stringer cobbled or cocured to the skin. The geometry is as shown on next figure.

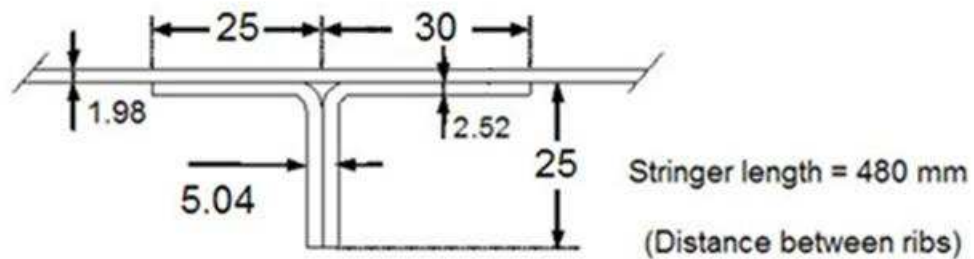


Figure 11. Example 3 Geometry

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ϵ_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

- Lay-up: Stringer (+45/-45/0/0/0/90/0)S
Skin (45/-45/0/0/90/0)\$

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = 1.98 \text{ mm}$$

$$t_t = 2.52 + 1.98 = 4.50 \text{ mm}$$

Since $t_t \geq 2 \cdot t_{sk}$, three segments are going to be considered.

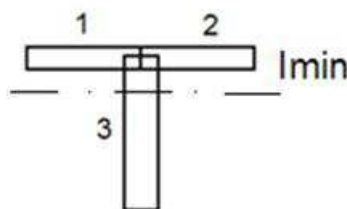


Figure 12. Example 3 : Segments

The dimensions and lay-up to be considered for these segments are:

Segment	W (mm)	t (mm)	Lay-up
1	25	4.5	45/-45/0/-45/45/90/-45/45/0/45/-45/90/45/-45/0/-45/45/ 45/-45/0/0/0/90/0/0/90/0/0/0/-45/45
2	25	4.5	45/-45/0/-45/45/90/-45/45/0/45/-45/90/45/-45/0/-45/45/ 45/-45/0/0/0/90/0/0/90/0/0/0/-45/45
3	$25+1.98-4.5/2=44.73$	5.04	(+45/-45/0/0/0/90/0/0/90/0/0/0/-45/+45)S

Therefore, segments 1 and 2 are parallels to the direction of minor inertia, $I_p = 1$. Meanwhile, the vertical segment number 3 is perpendicular to the direction of minor inertia, $I_p = 2$.

For all the segments it has one end free and the other en supported, $I_s = 1$.

The material properties required by the program for the laminate of the considered segments can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

Segment	1 & 2	3
E_c	73937.5 N/mm ²	79410.4 N/mm ²
D_{11}	546605.6 N/mm ²	825340.3 N/mm ²
D_{12}	104087 N/mm ²	133448.9 N/mm ²
D_{22}	267552.7 N/mm ²	344068.0 N/mm ²
D_{33}	118540.7 N/mm ²	153754.0 N/mm ²
ϵ_c	10000.0 $\mu\epsilon$	10000.0 $\mu\epsilon$

The inertia properties are obtained by running AIDA program with thickness normalized with the Young Modulus of segment 1.

Segment	E_c (N/mm ²)	th_{real} (mm)	th_{norm} (mm)
1	73937.5	4.5	4.5
2	73937.5	4.5	4.5
3	79410.4	5.04	5.4

The results are shown in figure below.

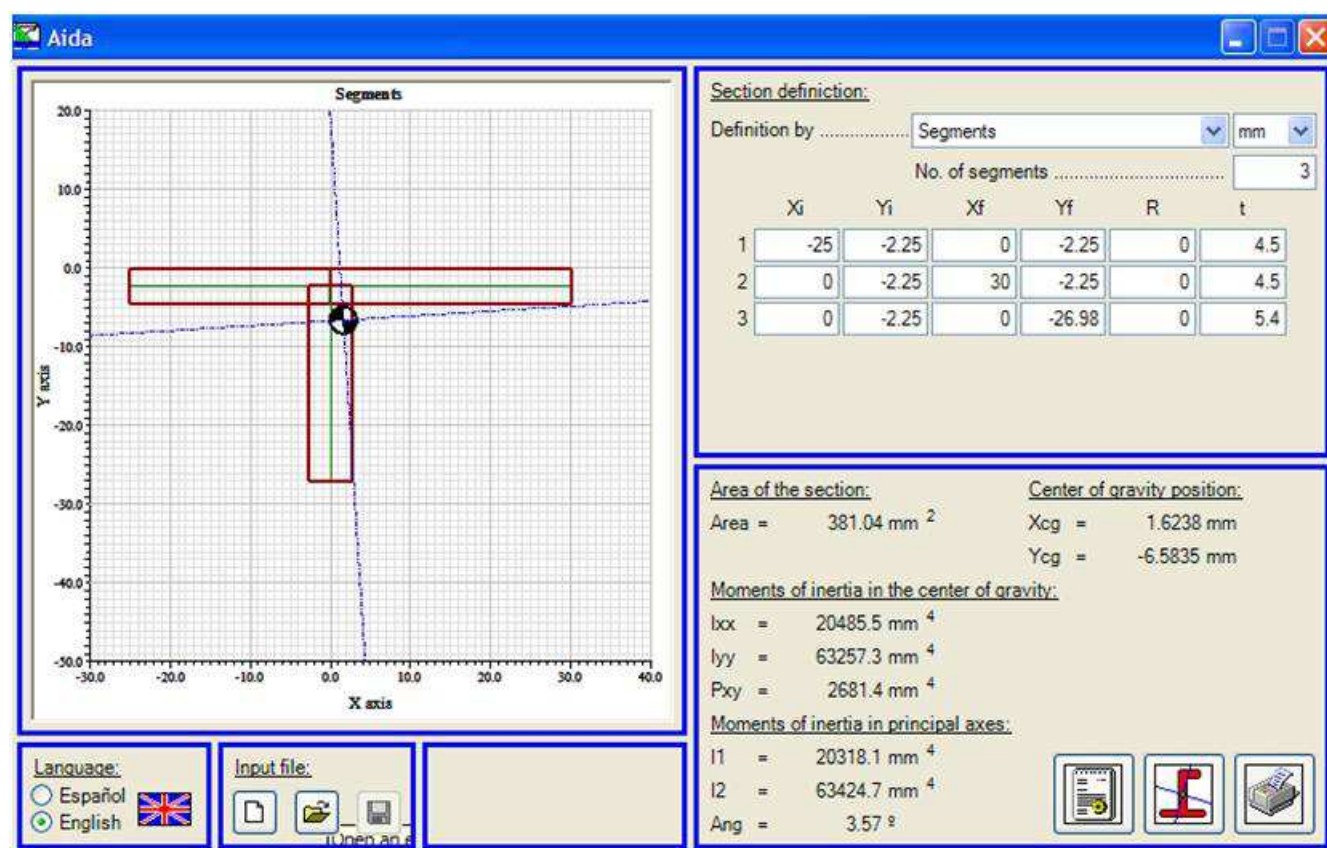


Figure 13. Example 3 Inertia Properties

The input and output files corresponding to this example are shown on next pages:

8.3.1 Input file Example 3:

```

1 Program CRIPCO Example of utilization 3
  Stringer in "T" of 480 mm length
  Section with three segments
  Vertical segment perpendicular direction minor inertia
2 Length
  480.0
3 Number of Segments
  3
4 Segment data: Orient.  Apoyo  Anchura  Espesor  Ec  D11  D12  D22  D33  Ecu
  1 1   25.00   4.500  73937.5  546605.6  104087.0  267552.7  118540.7  10000.0
  1 1   30.00   4.500  73937.5  546605.6  104087.0  267552.7  118540.7  10000.0
  2 1   24.73   5.040  79410.4  825340.3  133448.9  344068.0  153754.0  10000.0
  
```

8.3.2 Output file Example 3

```
*****
*
*          <<< CRIPCO >>>
*
*****
* Calculation of the crippling allowable
* value for composite stringers
*****
```

Title:
=====

Program CRIPCO Example of utilization 3
 Stringer in "T" of 480 mm length
 Section with three segments
 Vertical segment perpendicular direction minor inertia

Input data:
=====

Length of the stiffener: 480.0

Number of stiffener segments: 3

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	25.00	4.500	1	1	73937.5	546605.6	104087.0	267552.7	118540.7	10000.0
2	30.00	4.500	1	1	73937.5	546605.6	104087.0	267552.7	118540.7	10000.0
3	24.73	5.040	2	1	79410.4	825340.3	133448.9	344068.0	153754.0	10000.0

Allowable values for local buckling and crippling:
 =====

Segment	Local buckling	Crippling
1	-7122.1	-7917.7
2	-5031.9	-7096.5
3	-7891.3	-7917.7

```
*****
* Crippling microstrains for the stiffener: -7627.0 *
*****
```

8.4 Example 4

The case to solve is a I-shape stringer cobonded or cocured to the skin. The geometry is as shown on next figure.

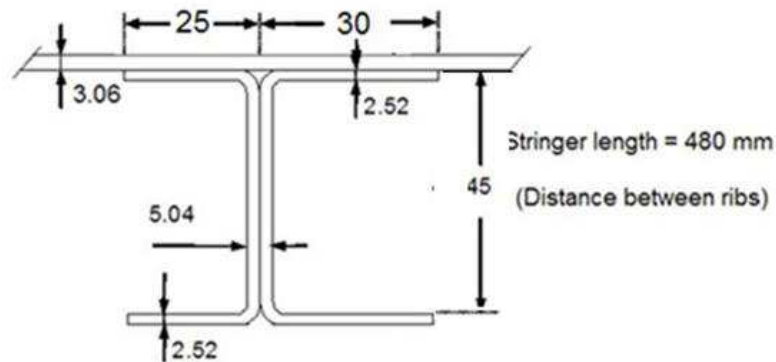


Figure 14. Example 4 Geometry

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ϵ_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

- Lay-up: Stringer (+45/-45/0/0/0/90/0)S
Skin (45/-45/0/0/0/90/0)\$

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = 3.06 \text{ mm}$$

$$t_t = 2.52 + 3.06 = 5.58 \text{ mm}$$

Since $t_t < 2 \cdot t_{sk}$, three segments are going to be considered.

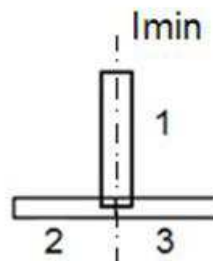


Figure 15. Example 4 : Segments

The dimensions and lay-up to be considered for these segments are:

Segment	W (mm)	t (mm)	Lay-up
1	$45+3.06-5.58/2+2.52/2=46.53$	5.04	(+45/-45/0/0/0/90/0/0/90/0/0/0/-45/+45)S
2	25	2.52	(45/-45/0/0/0/90/0)S
3	25	2.52	(45/-45/0/0/0/90/0)S

The material properties required by the program for the laminate of the considered segments can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

Segment	1	2 & 3
E_c	79410.4 N/mm ²	79410.4 N/mm ²
D_{11}	825340.3 N/mm ²	87988.4 N/mm ²
D_{12}	133448.9 N/mm ²	25975.1 N/mm ²
D_{22}	344068.0 N/mm ²	39599.6 N/mm ²
D_{33}	153754.0 N/mm ²	28513.3 N/mm ²
ϵ_c	10000.0 $\mu\epsilon$	10000.0 $\mu\epsilon$

The inertia properties are obtained by running AIDA program with thickness normalized with the Young Modulus of segment 1.

Segment	E_c (N/mm ²)	th_{real} (mm)	th_{norm} (mm)
1	79410.4	5.04	5.04
2	79410.4	2.52	2.52
3	79410.4	2.52	2.52

The results are shown in figure below.

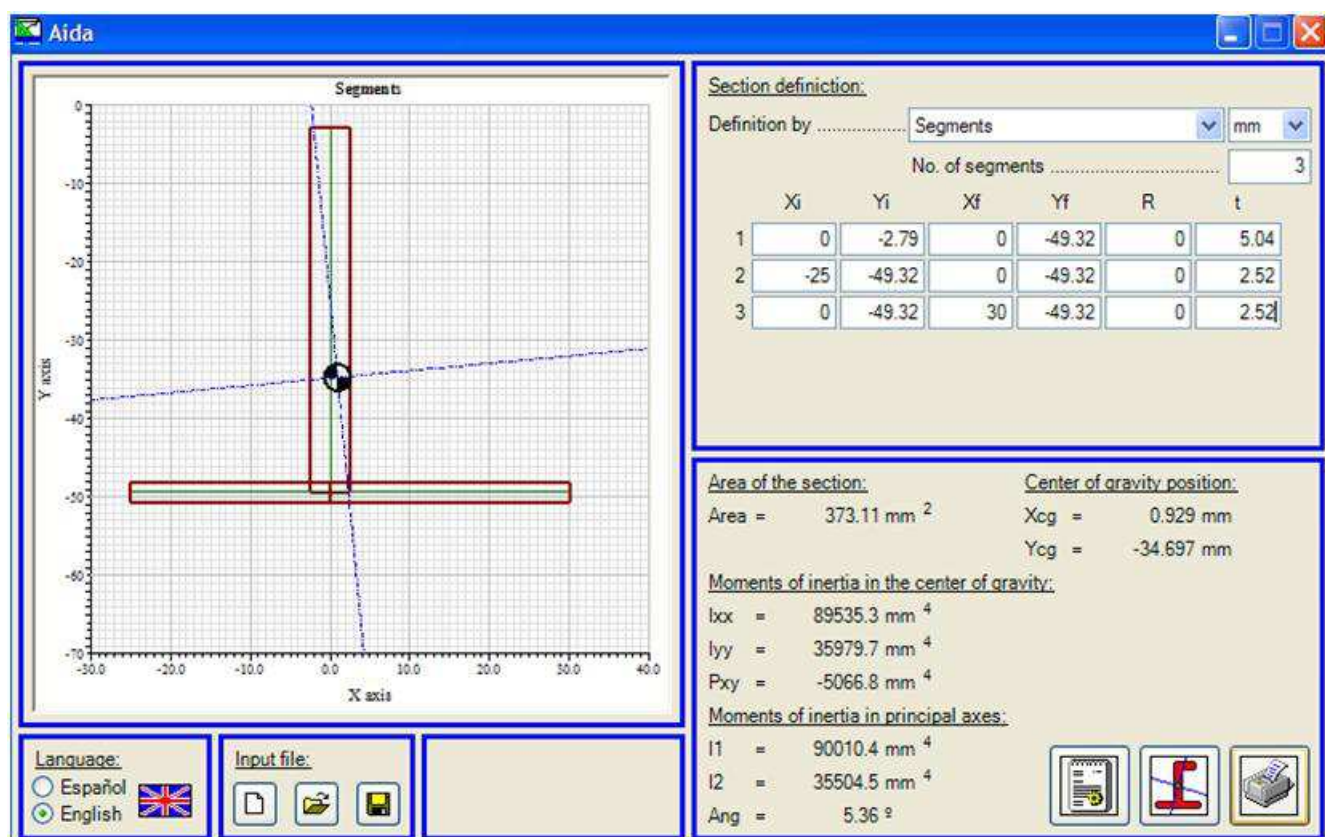


Figure 16. Example 4 Inertia Properties

Therefore, vertical segment number 1 is parallel to the direction of minor inertia, $I_p = 1$. Meanwhile, the segments number 2 and 3 are perpendiculars to the direction of minor inertia, $I_p = 2$.

Vertical segment has the two end supported, $I_s = 2$. Meanwhile, horizontal segments have one end free and the other end supported, $I_s = 1$.

The input and output files corresponding to this example are shown on next pages:

8.4.1 Input file Example 4:

```

1 Program CRIPCO Example of utilization 4
  Stringer in "I" of 480 mm length
  Section with three segments
  Vertical segment parallel direction minor inertia
2 Length
  480.0
3 Number of segments
  3
4 Segment data: Orient.  Apoyo  Anchura  Espesor  Ec  D11  D12  D22  D33  Ecu
  1 2   46.53   5.040  79410.4   825340.3  133448.9  344068.0  153754.0  10000.0
  2 1   25.00   2.520  79410.4   87988.4   25975.1   39559.6   28513.3   10000.0
  2 1   30.00   2.520  79410.4   87988.4   25975.1   39559.6   28513.3   10000.0
  
```


8.4.2 Output file Example 4

```
*****
*
*          <<< CRIPCO >>>
*
*****
* Calculation of the crippling allowable
* value for composite stringers
*****
```

Title:
=====

Program CRIPCO Example of utilization 4
 Stringer in "I" of 480 mm length
 Section with three segments
 Vertical segment parallel direction minor inertia

Input data:
=====

Length of the stiffener: 480.0

Number of stiffener segments: 3

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	46.53	5.040	1	2	79410.4	825340.3	133448.9	344068.0	153754.0	10000.0
2	25.00	2.520	2	1	79410.4	87988.4	25975.1	39559.6	28513.3	10000.0
3	30.00	2.520	2	1	79410.4	87988.4	25975.1	39559.6	28513.3	10000.0

Allowable values for local buckling and crippling:
=====

Segment	Local buckling	Crippling
1	-22184.4	-5284.8
2	-2811.1	-5284.8
3	-1975.1	-5284.8

```
*****
* Crippling microstrains for the stiffener: -5284.8 *
*****
```

8.5 Example 5

The case to solve is a I-shape stringer cobonded or cocured to the skin. The geometry is as shown on next figure.

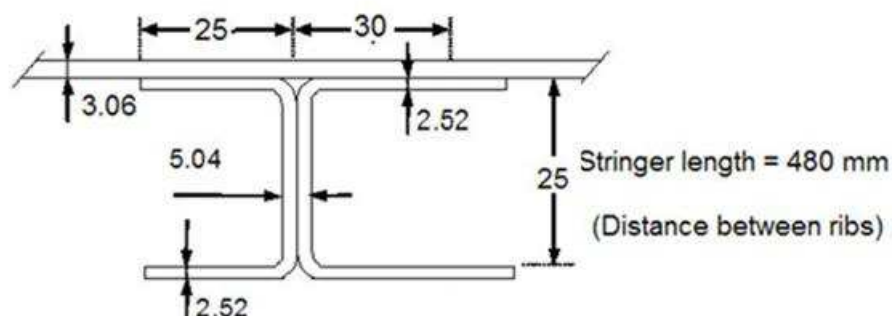


Figure 17. Example 5 Geometry

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ϵ_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

- Lay-up: Stringer (+45/-45/0/0/0/90/0)S
Skin (45/-45/0/0/90/0)\$

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = 3.06 \text{ mm}$$

$$t_t = 2.52 + 3.06 = 5.58 \text{ mm}$$

Since $t_t < 2 \cdot t_{sk}$, three segments are going to be considered.

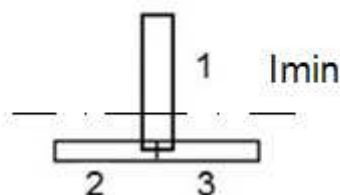


Figure 18. Example 5 : Segments

The dimensions and lay-up to be considered for these segments are:

Segment	W (mm)	t (mm)	Lay-up
1	$25+3.06-5.58/2+2.52/2=26.53$	5.04	(+45/-45/0/0/0/90/0/0/90/0/0/0/-45/+45)S
2	25	2.52	(45/-45/0/0/0/90/0)S
3	25	2.52	(45/-45/0/0/0/90/0)S

The material properties required by the program for the laminate of the considered segments can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

Segment	1	2 & 3
E_c	79410.4 N/mm ²	73937.5 N/mm ²
D_{11}	825340.3 N/mm ²	87988.4 N/mm ²
D_{12}	133448.9 N/mm ²	25975.1 N/mm ²
D_{22}	344068.0 N/mm ²	39599.6 N/mm ²
D_{33}	153754.0 N/mm ²	28513.3 N/mm ²
ϵ_c	10000.0 $\mu\epsilon$	10000.0 $\mu\epsilon$

The inertia properties are obtained by running AIDA program with thickness normalized with the Young Modulus of segment 1.

Segment	E_c (N/mm ²)	th_{real} (mm)	th_{norm} (mm)
1	79410.4	5.04	5.04
2	79410.4	2.52	2.52
3	79410.4	2.52	2.52

The results are shown in figure below.

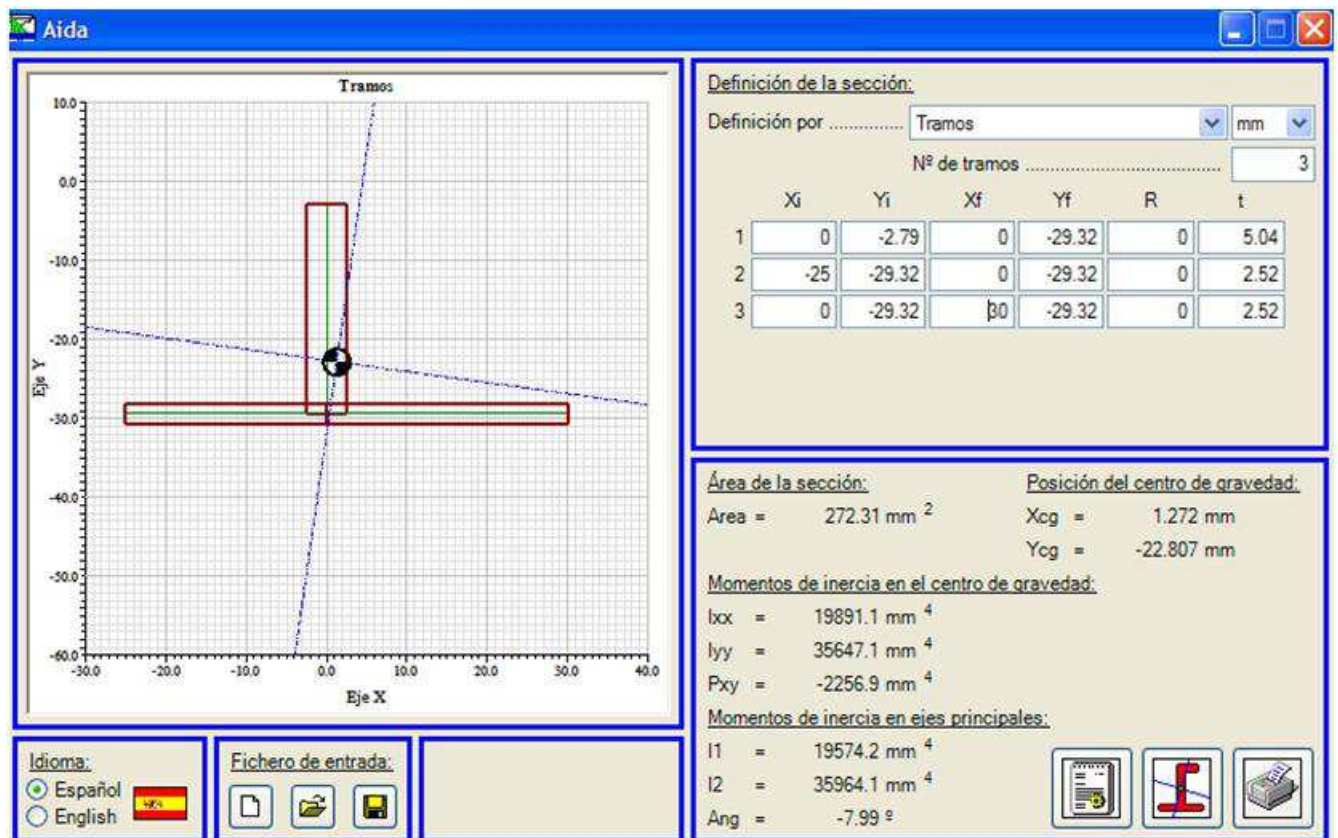


Figure 19. Example 5 Inertia Properties

Therefore, vertical segment number 1 is parallel to the direction of minor inertia, $I_p = 1$. Meanwhile, the segments number 2 and 3 are perpendicular to the direction of minor inertia, $I_p = 2$.

Vertical segment has the two end supported, $I_s = 2$. Meanwhile, horizontal segments have one end free and the other end supported, $I_s = 1$.

The input and output files corresponding to this example are shown on next pages:

8.5.1 Input file Example 5:

```

1 Program CRIPCO Example of utilization 5
  Stringer in "I" of 480 mm length
  Section with three segments
  Vertical segment perpendicular direction minor inertia
2 Longitud
  480.0
3 Número de tramos
  3
4 Datos de tramo: Orient.  Apoyo  Anchura  Espesor  Ec  D11  D12  D22  D33  Ecu
  2  2   26.53   5.040  79410.4   825340.3   133448.9   344068.0   153754.0   10000.0
  1  1   25.00   2.520  79410.4   87988.4   25975.1   39559.6   28513.3   10000.0
  1  1   30.00   2.520  79410.4   87988.4   25975.1   39559.6   28513.3   10000.0
  
```

8.5.2 Output file Example 5

```
*****
*
*          <<< CRIPCO >>>
*
*****
* Calculation of the crippling allowable
* value for composite stringers
*****
```

Title:
=====

Program CRIPCO Example of utilization 5
 Stringer in "I" of 480 mm length
 Section with three segments
 Vertical segment perpendicular direction minor inertia

Input data:
=====

Length of the stiffener: 480.0

Number of stiffener segments: 3

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	26.53	5.040	2	2	79410.4	825340.3	133448.9	344068.0	153754.0	10000.0
2	25.00	2.520	1	1	79410.4	87988.4	25975.1	39559.6	28513.3	10000.0
3	30.00	2.520	1	1	79410.4	87988.4	25975.1	39559.6	28513.3	10000.0

Allowable values for local buckling and crippling:
=====

Segment	Local buckling	Crippling
1	-68239.9	-5906.6
2	-2811.1	-5906.6
3	-1975.1	-5284.8

```
*****
* Crippling microstrains for the stiffener: -5734.0 *
*****
```

8.6 Example 6

The case to solve is a I-shape stringer cobonded or cocured to the skin. The geometry is as shown on next figure.

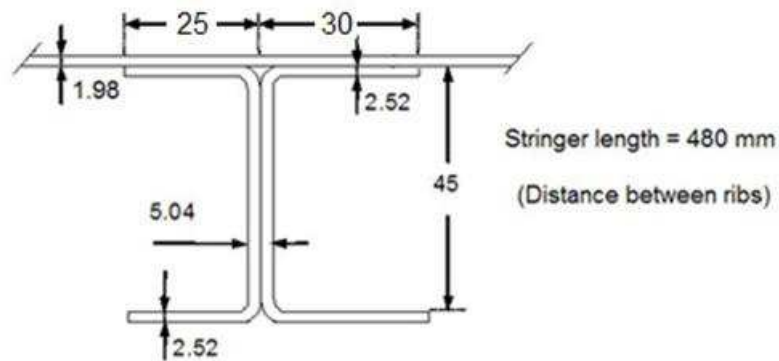


Figure 20. Example 6 Geometry

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ϵ_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

- Lay-up: Stringer (+45/-45/0/0/90/0)S
 Skin (45/-45/0/0/90/0)\$

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = 1.98 \text{ mm}$$

$$t_t = 2.52 + 1.98 = 4.5 \text{ mm}$$

Since $t_t \geq 2 \cdot t_{sk}$, five segments are going to be considered.

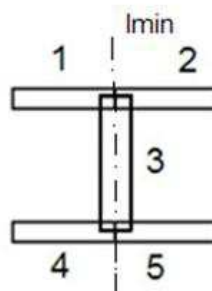


Figure 21. Example 6 : Segments

The dimensions and lay-up to be considered for these segments are:

Segment	W (mm)	t (mm)	Lay-up
1	25	4.5	45/-45/0/-45/45/90/-45/45/0/45/-45/90/45/-45/0/-45/45/ 45/-45/0/0/0/90/0/0/90/0/0/0/-45/45
2	25	4.5	45/-45/0/-45/45/90/-45/45/0/45/-45/90/45/-45/0/-45/45/ 45/-45/0/0/0/90/0/0/90/0/0/0/-45/45
3	$45+1.98-4.5/2+2.52/2=45.99$	5.04	(+45/-45/0/0/0/90/0/0/90/0/0/0/-45/+45)S
4	25	2.52	(45/-45/0/0/0/90/0)S
5	25	2.52	(45/-45/0/0/0/90/0)S

The material properties required by the program for the laminate of the considered segments can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

Segment	1 & 2	3	4 & 5
E_c	73937.5 N/mm ²	79410.4 N/mm ²	79410.4 N/mm ²
D_{11}	546605.6 N/mm ²	825340.3 N/mm ²	87988.4 N/mm ²
D_{12}	104087 N/mm ²	133448.9 N/mm ²	25975.1 N/mm ²
D_{22}	267552.7 N/mm ²	344068.0 N/mm ²	39599.6 N/mm ²
D_{33}	118540.7 N/mm ²	153754.0 N/mm ²	28513.3 N/mm ²
ϵ_c	10000.0 $\mu\epsilon$	10000.0 $\mu\epsilon$	10000.0 $\mu\epsilon$

The inertia properties are obtained by running AIDA program with thickness normalized with Young Modulus of segment1.

Segment	E_c (N/mm ²)	th_{real} (mm)	th_{norm} (mm)
1	73937.5	4.5	4.5
2	73937.5	4.5	4.5
3	79410.4	5.04	5.4
1	79410.4	2.52	2.7
2	79410.4	2.52	2.7

The results are shown in figure below.

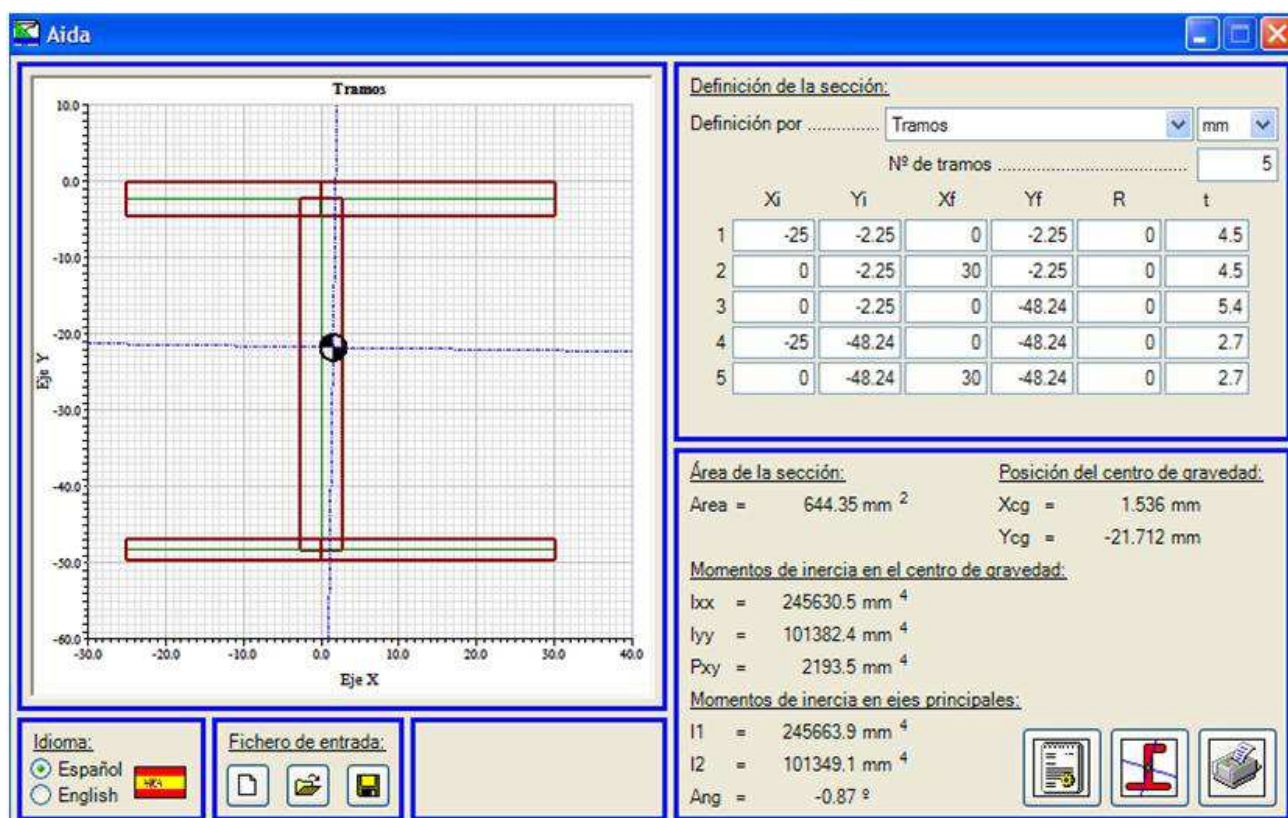


Figure 22. Example 6 Inertia Properties

Therefore, vertical segment number 3 is parallel to the direction of minor inertia, $I_p = 1$. Meanwhile, the horizontal segments are perpendicular to the direction of minor inertia, $I_p = 2$.

For all the horizontal segments it has one end free and the other en supported, $I_s = 1$. The vertical segment number 3 has the two end supported, $I_s = 2$

The input and output files corresponding to this example are shown on next pages:

8.6.1 Input file Example 6:

```

1 Program CRIPCO Example of utilization 6
  Stringer in "I" of 480 mm length
  Section with five segments
  Vertical segment pearallel direction minor inertia
2 Length
  480.0
3 Number of segments
  5
4 Segment data: Orient.  Apoyo  Anchura  Espesor  Ec  D11  D12  D22  D33  Ecu
  2  1  25.00  4.500  73937.5  546605.6  104087.0  267552.7  118540.7  10000.0
  2  1  30.00  4.500  73937.5  546605.6  104087.0  267552.7  118540.7  10000.0
  1  2  45.99  5.040  79410.4  825340.3  133448.9  344068.0  153754.0  10000.0
  2  1  25.00  2.520  79410.4  87988.4  25975.1  39559.6  28513.3  10000.0
  2  1  30.00  2.520  79410.4  87988.4  25975.1  39559.6  28513.3  10000.0

```

8.6.2 Output file Example 6

```
*****
*
*          <<< CRIPCO >>>
*
*****
* Calculation of the crippling allowable
* value for composite stringers
*****
```

Title:
=====

Program CRIPCO Example of utilization 6
 Stringer in "I" of 480 mm length
 Section with five segments
 Vertical segment pearallel direction minor inertia

Input data:
=====

Length of the stiffener: 480.0

Number of stiffener segments: 5

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	25.00	4.500	2	1	73937.5	546605.6	104087.0	267552.7	118540.7	10000.0
2	30.00	4.500	2	1	73937.5	546605.6	104087.0	267552.7	118540.7	10000.0
3	45.99	5.040	1	2	79410.4	825340.3	133448.9	344068.0	153754.0	10000.0
4	25.00	2.520	2	1	79410.4	87988.4	25975.1	39559.6	28513.3	10000.0
5	30.00	2.520	2	1	79410.4	87988.4	25975.1	39559.6	28513.3	10000.0

Allowable values for local buckling and crippling:

```
=====
```

Segment	Local buckling	Crippling
1	-7122.1	-5284.8
2	-5031.9	-5284.8
3	-22708.4	-5284.8
4	-2811.1	-5284.8
5	-1975.1	-5284.8

```
*****
* Crippling microstrains for the stiffener: -5284.8 *
*****
```

8.7 Example 7

The case to solve is a I-shape stringer cobonded or cocured to the skin. The geometry is as shown on next figure.

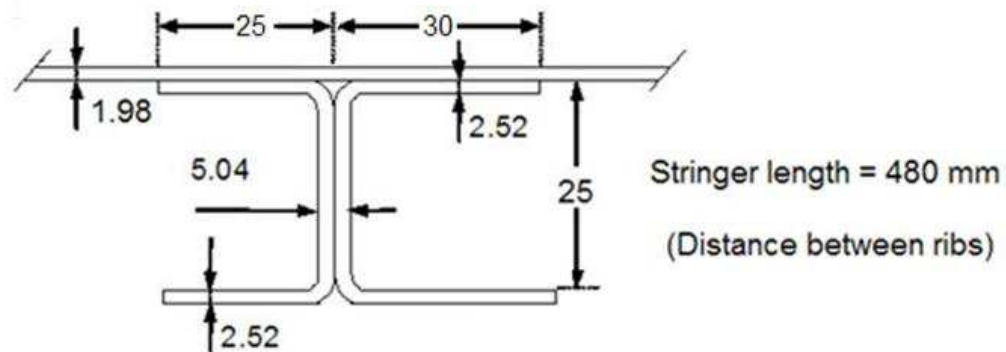


Figure 23. Example 7 Geometry

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ϵ_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

- Lay-up: Stringer (+45/-45/0/0/0/90/0)S
Skin (45/-45/0/0/90/0)\$

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = 1.98 \text{ mm}$$

$$t_t = 2.52 + 1.98 = 4.5 \text{ mm}$$

Since $t_t \geq 2 \cdot t_{sk}$, five segments are going to be considered.

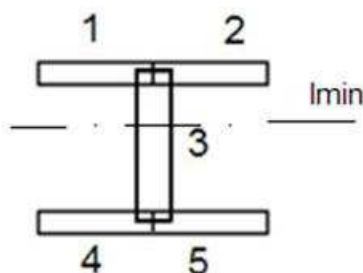


Figure 24. Example 7 : Segments

The dimensions and lay-up to be considered for these segments are:

Segment	W (mm)	t (mm)	Lay-up
1	25	4.5	45/-45/0/-45/45/90/-45/45/0/45/-45/90/45/-45/0/-45/45/ 45/-45/0/0/0/90/0/0/90/0/0/0/-45/45
2	25	4.5	45/-45/0/-45/45/90/-45/45/0/45/-45/90/45/-45/0/-45/45/ 45/-45/0/0/0/90/0/0/90/0/0/0/-45/45
3	$25+1.98-4.5/2+2.52/2=25.99$	5.04	(+45/-45/0/0/0/90/0/0/90/0/0/0/-45/+45)S
4	25	2.52	(45/-45/0/0/0/90/0)S
5	25	2.52	(45/-45/0/0/0/90/0)S

The material properties required by the program for the laminate of the considered segments can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

Segment	1 & 2	3	4 & 5
E_c	73937.5 N/mm ²	79410.4 N/mm ²	73937.5 N/mm ²
D_{11}	546605.6 N/mm ²	825340.3 N/mm ²	87988.4 N/mm ²
D_{12}	104087 N/mm ²	133448.9 N/mm ²	25975.1 N/mm ²
D_{22}	267552.7 N/mm ²	344068.0 N/mm ²	39599.6 N/mm ²
D_{33}	118540.7 N/mm ²	153754.0 N/mm ²	28513.3 N/mm ²
ϵ_c	10000.0 $\mu\epsilon$	10000.0 $\mu\epsilon$	10000.0 $\mu\epsilon$

The inertia properties are obtained by running AIDA program with thickness normalized with Young Modulus of segment1.

Segment	E_c (N/mm ²)	th_{real} (mm)	th_{norm} (mm)
1	73937.5	4.5	4.5
2	73937.5	4.5	4.5
3	79410.4	5.04	5.4
1	79410.4	2.52	2.7
2	79410.4	2.52	2.7

The results are shown in figure below.

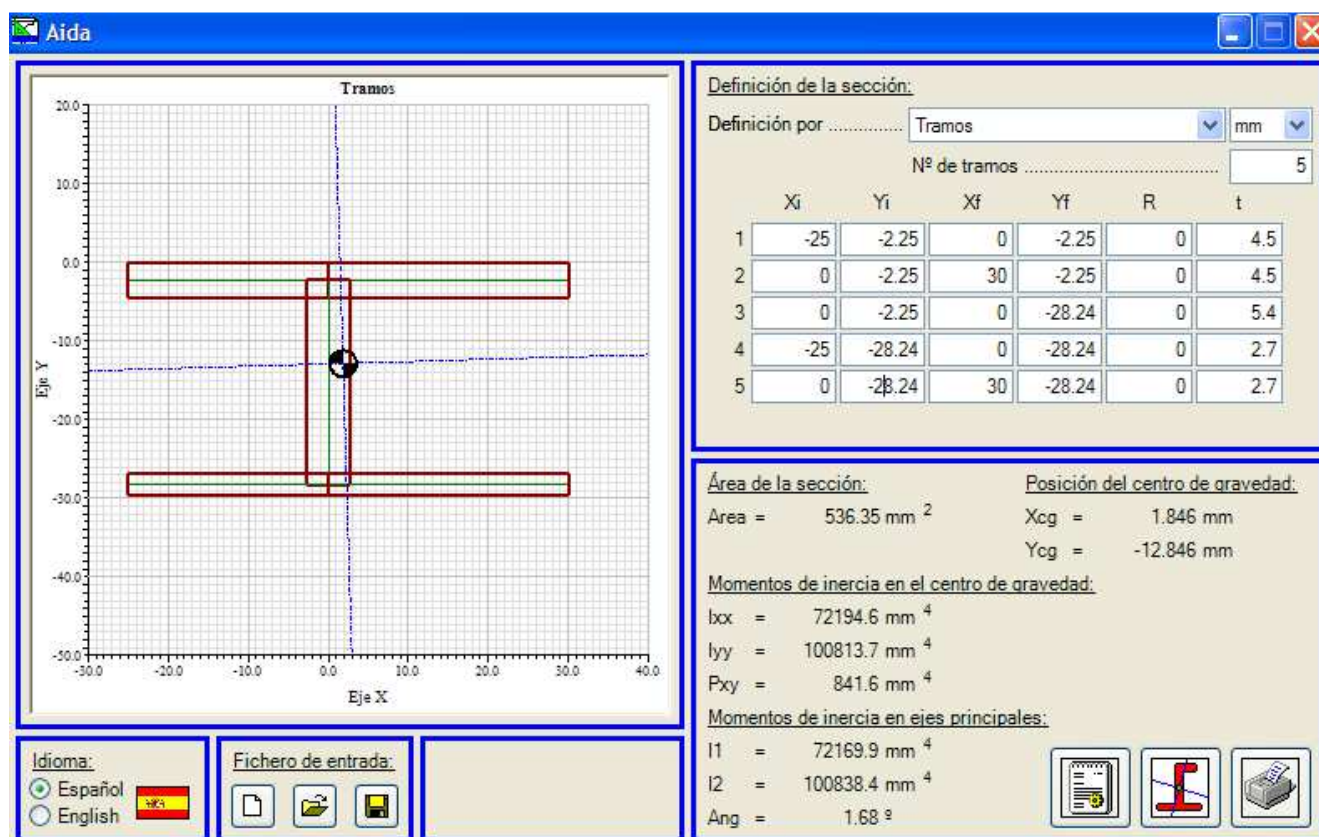


Figure 25. Example 7 Inertia Properties

Therefore, vertical segment number 3 is perpendicular to the direction of minor inertia, $I_p = 2$. Meanwhile, the horizontal segments are parallel to the direction of minor inertia, $I_p = 1$.

For all the horizontal segments it has one end free and the other en supported, $I_s = 1$. The vertical segment number 3 has the two end supported, $I_s = 2$

The input and output files corresponding to this example are shown on next pages:

8.7.1 Input file Example 7:

```

1 Program CRIPCO Example of utilization 7
  Stringer in"I" of 480 mm length
  Section with five segments
  Vertical segment perpendicular direction minor inertia
2 Length
  480.0
3 Number of segments
  5
4 Segment data: Orient.  Apoyo  Anchura  Espesor  Ec  D11  D12  D22  D33  Ecu
  1 1  25.00  4.500  73937.5  546605.6  104087.0  267552.7  118540.7  10000.0
  1 1  30.00  4.500  73937.5  546605.6  104087.0  267552.7  118540.7  10000.0
  2 2  25.99  5.040  79410.4  825340.3  133448.9  344068.0  153754.0  10000.0
  1 1  25.00  2.520  79410.4  87988.4  25975.1  39559.6  28513.3  10000.0
  1 1  30.00  2.520  79410.4  87988.4  25975.1  39559.6  28513.3  10000.0

```

8.7.2 Output file Example 7

* *
 * <<< CRIPCO >>> *
 * *

* Calculation of the crippling allowable *
 * value for composite stringers *

Title:

=====

Program CRIPCO Example of utilization 7
 Stringer in "I" of 480 mm length
 Section with five segments
 Vertical segment perpendicular direction minor inertia

Input data:

=====

Length of the stiffener: 480.0

Number of stiffener segments: 5

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	25.00	4.500	1	1	73937.5	546605.6	104087.0	267552.7	118540.7	10000.0
2	30.00	4.500	1	1	73937.5	546605.6	104087.0	267552.7	118540.7	10000.0
3	25.99	5.040	2	2	79410.4	825340.3	133448.9	344068.0	153754.0	10000.0
4	25.00	2.520	1	1	79410.4	87988.4	25975.1	39559.6	28513.3	10000.0
5	30.00	2.520	1	1	79410.4	87988.4	25975.1	39559.6	28513.3	10000.0

Allowable values for local buckling and crippling:

=====

Segment	Local buckling	Crippling
1	-7122.1	-5906.6
2	-5031.9	-5906.6
3	-71105.1	-5906.6
4	-2811.1	-5906.6
5	-1975.1	-5284.8

* Crippling microstrains for the stiffener: -5812.6 *

8.8 Example 8

The case to solve is a Z-shape stringer cobonded or cocured to the skin. The geometry is as shown on next figure.

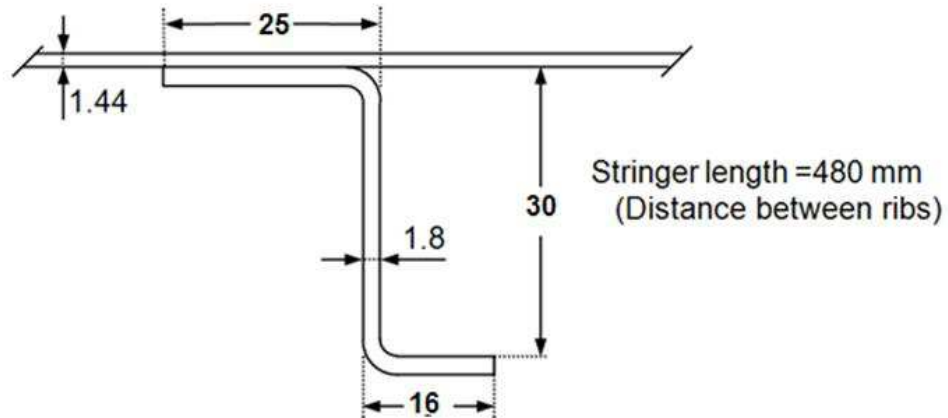


Figure 26. Example 8 geometry.

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ϵ_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

- Lay-up: Stringer (90/45/0/0/-45)S
Skin (90/45/0/-45)S

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = 1.44 \text{ mm}$$

$$t_t = 1.8 + 1.44 = 3.24 \text{ mm}$$

Since $t_t > 2 \cdot t_{sk}$, three segments are going to be considered.

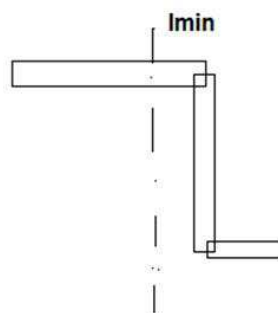


Figure 27. Example 8 : Segments

The dimensions of these three segments are:

$$W_{upper} = 25 - 1.8/2 = 24.1 \text{ mm}$$

$$t_{upper} = 1.8 + 1.44 = 3.24 \text{ mm}$$

$$W_{web} = 31.44 - (1.8 + 3.24)/2 = 28.92 \text{ mm}$$

$$t_{web} = 1.8 \text{ mm}$$

$$W_{lower} = 16 - 1.8/2 = 15.1 \text{ mm}$$

$$t_{lower} = 1.8 \text{ mm}$$

The material properties required by the program for the laminate can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

Upper flange + Skin

$$E_c = 54932.1 \text{ N/mm}^2$$

$$D_{11} = 148333.6 \text{ N/mm}^2$$

$$D_{12} = 38753.2 \text{ N/mm}^2$$

$$D_{22} = 155744.4 \text{ N/mm}^2$$

$$D_{33} = 44147.6 \text{ N/mm}^2$$

$$\epsilon_c = 10000.0 \text{ } \mu\epsilon$$

Lower flange and web

$$E_c = 62374.4 \text{ N/mm}^2$$

$$D_{11} = 19899.7 \text{ N/mm}^2$$

$$D_{12} = 5186.2 \text{ N/mm}^2$$

$$D_{22} = 35157.4 \text{ N/mm}^2$$

$$D_{33} = 6111.1 \text{ N/mm}^2$$

$$\epsilon_c = 10000.0 \text{ } \mu\epsilon$$

The inertia properties are obtained by running AIDA program with thickness normalized with Young Modulus of segment1.

Segment	E_c (N/mm ²)	th_{real} (mm)	th_{norm} (mm)
1	54932.1	3.24	3.24
2	62374.4	1.8	2.04
3	62374.4	1.8	2.04

The results are shown in figure below

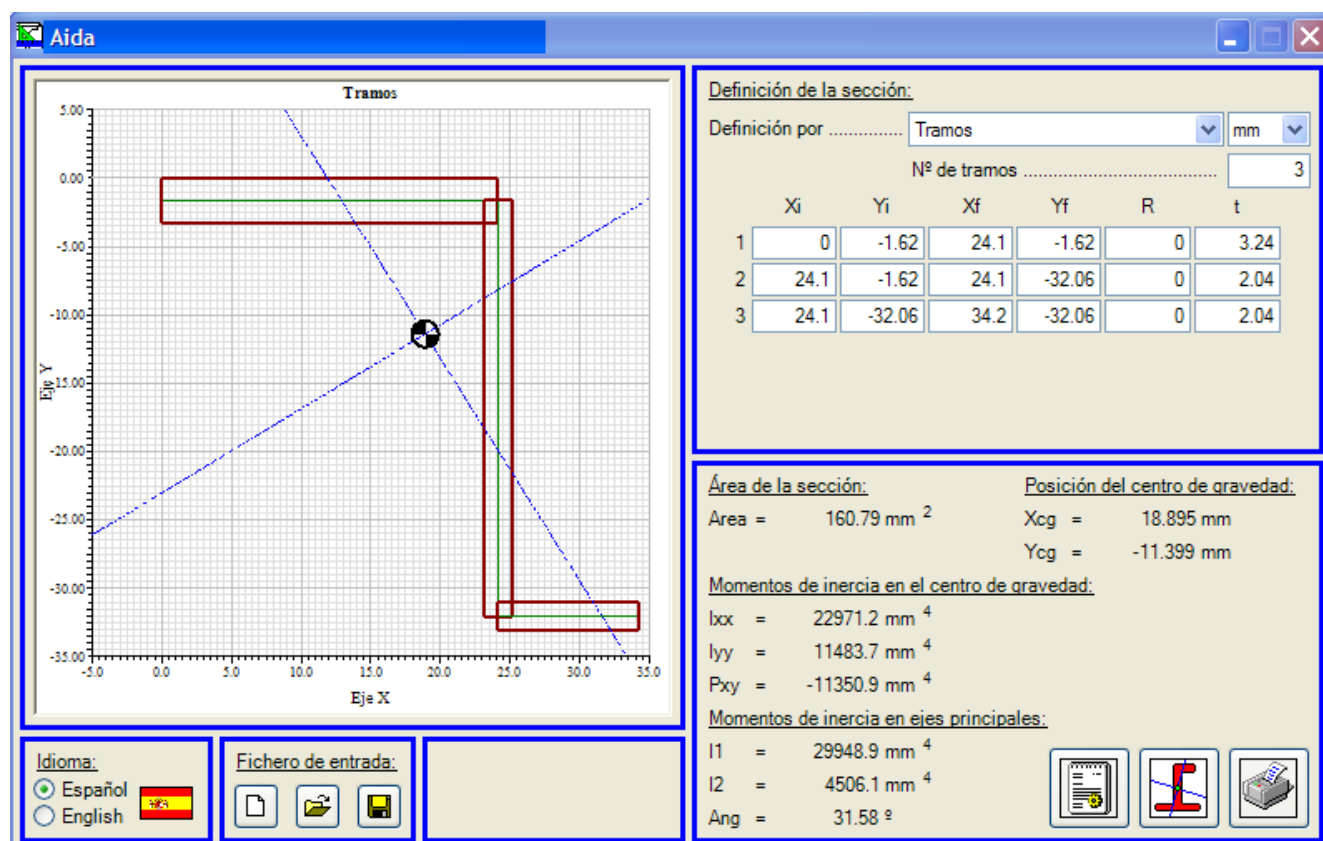


Figure 28. Example 8 Inertia properties

The upper and lower flanges are perpendicular to the direction of minor inertia, $I_p = 2$, and they have one end free and the other supported, $I_s = 1$. The web of the stringer is parallel to the direction of minor inertia, $I_p = 1$ and it has both ends supported, $I_s = 2$.

The input and output files corresponding to this example are shown on next pages:

8.8.1 Input file Example 8:

```

1 Program CRIPCO Example of utilization 8
  Stringer in "Z" of 480 mm length
  Section with three segments
  Vertical segment parallel direction minor inertia
2 Length
  480.0
3 Number of segments
  3
4 Segment data: Orient.  Apoyo  Anchura  Espesor  Ec  D11  D12  D22  D33  Ecu
  2  1   24.10   3.240  54932.1  148333.6  38753.2  155744.4  44147.6  10000.0
  1  2   28.92   1.800  62374.4  19899.7   5186.2   35157.4   6111.1  10000.0
  2  1   15.10   1.800  62374.4  19899.7   5186.2   35157.4   6111.1  10000.0
  
```

8.8.2 Output file Example 8

```
*****
*                                     *
*          <<< CRIPCO >>>          *
*                                     *
*****
* Calculation of the crippling allowable *
* value for composite stringers       *
*****
```

Title:
=====

Program CRIPCO Example of utilization 8
 Stringer in "Z" of 480 mm length
 Section with three segments
 Vertical segment parallel direction minor inertia

Input data:
=====

Length of the stiffener: 480.0

Number of stiffener segments: 3

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	24.10	3.240	2	1	54932.1	148333.6	38753.2	155744.4	44147.6	10000.0
2	28.92	1.800	1	2	62374.4	19899.7	5186.2	35157.4	6111.1	10000.0
3	15.10	1.800	2	1	62374.4	19899.7	5186.2	35157.4	6111.1	10000.0

Allowable values for local buckling and crippling:
 =====

Segment	Local buckling	Crippling
1	-5267.7	-5961.7
2	-9219.6	-5961.7
3	-2895.0	-5961.7

```
*****
* Crippling microstrains for the stiffener: -5961.7 *
*****
```

8.9 Example 9

The case to solve is a Z-shape stringer cobonded or cocured to the skin. The geometry is as shown on next figure.

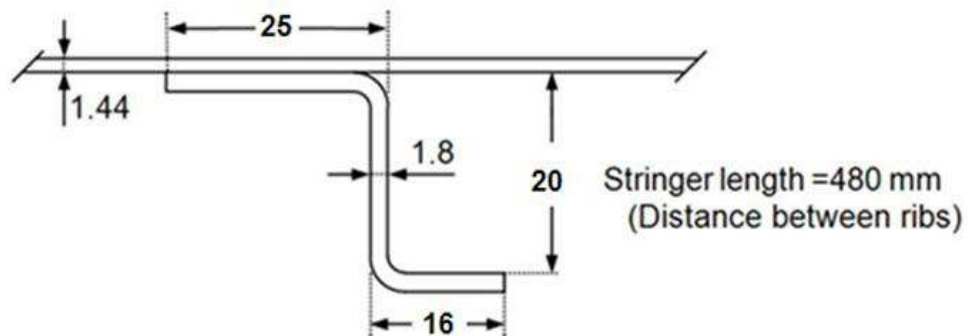


Figure 29. Example 9 geometry.

The stringer material is:

- Tape of group V:

The lamina properties under compression for this material are the following:

	E_{11}	E_{22}	G_{12}	μ_{12}	ϵ_c
Mean values	122000	8900	4700	0.3	---
Allowable values	120000	8600	4500	---	10000

- Lay-up: Stringer (90/45/0/0/-45)S
Skin (90/45/0/0/-45)S

The skin thickness and the total thickness of the stringer foot plus the skin are:

$$t_{sk} = 1.44 \text{ mm}$$

$$t_t = 1.8 + 1.44 = 3.24 \text{ mm}$$

Since $t_t > 2 \cdot t_{sk}$, three segments are going to be considered.

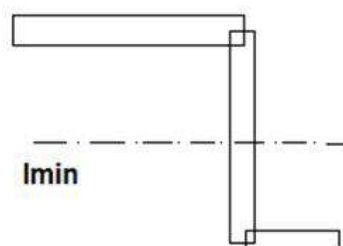


Figure 30. Example 8 : Segments

The dimensions of these three segments are:

$$W_{upper} = 25 - 1.8/2 = 24.1 \text{ mm}$$

$$t_{upper} = 1.8 + 1.44 = 3.24 \text{ mm}$$

$$W_{web} = 21.44 - (1.8 + 3.24)/2 = 18.92 \text{ mm}$$

$$t_{web} = 1.8 \text{ mm}$$

$$W_{lower} = 16 - 1.8/2 = 15.1 \text{ mm}$$

$$t_{lower} = 1.8 \text{ mm}$$

The material properties required by the program for the laminate can be calculated from the above data (i.e.: by running ANCOLAB program). The results are the following:

Upper flange + Skin

$$E_c = 54932.1 \text{ N/mm}^2$$

$$D_{11} = 148333.6 \text{ N/mm}^2$$

$$D_{12} = 38753.2 \text{ N/mm}^2$$

$$D_{22} = 155744.4 \text{ N/mm}^2$$

$$D_{33} = 44147.6 \text{ N/mm}^2$$

$$\epsilon_c = 10000.0 \text{ } \mu\epsilon$$

Lower flange and web

$$E_c = 62374.4 \text{ N/mm}^2$$

$$D_{11} = 19899.7 \text{ N/mm}^2$$

$$D_{12} = 5186.2 \text{ N/mm}^2$$

$$D_{22} = 35157.4 \text{ N/mm}^2$$

$$D_{33} = 6111.1 \text{ N/mm}^2$$

$$\epsilon_c = 10000.0 \text{ } \mu\epsilon$$

The inertia properties are obtained by running AIDA program with thickness normalized with Young Modulus of segment1.

Segment	E_c (N/mm ²)	th_{real} (mm)	th_{norm} (mm)
1	54932.1	3.24	3.24
2	62374.4	1.8	2.04
3	62374.4	1.8	2.04

The results are shown in figure below

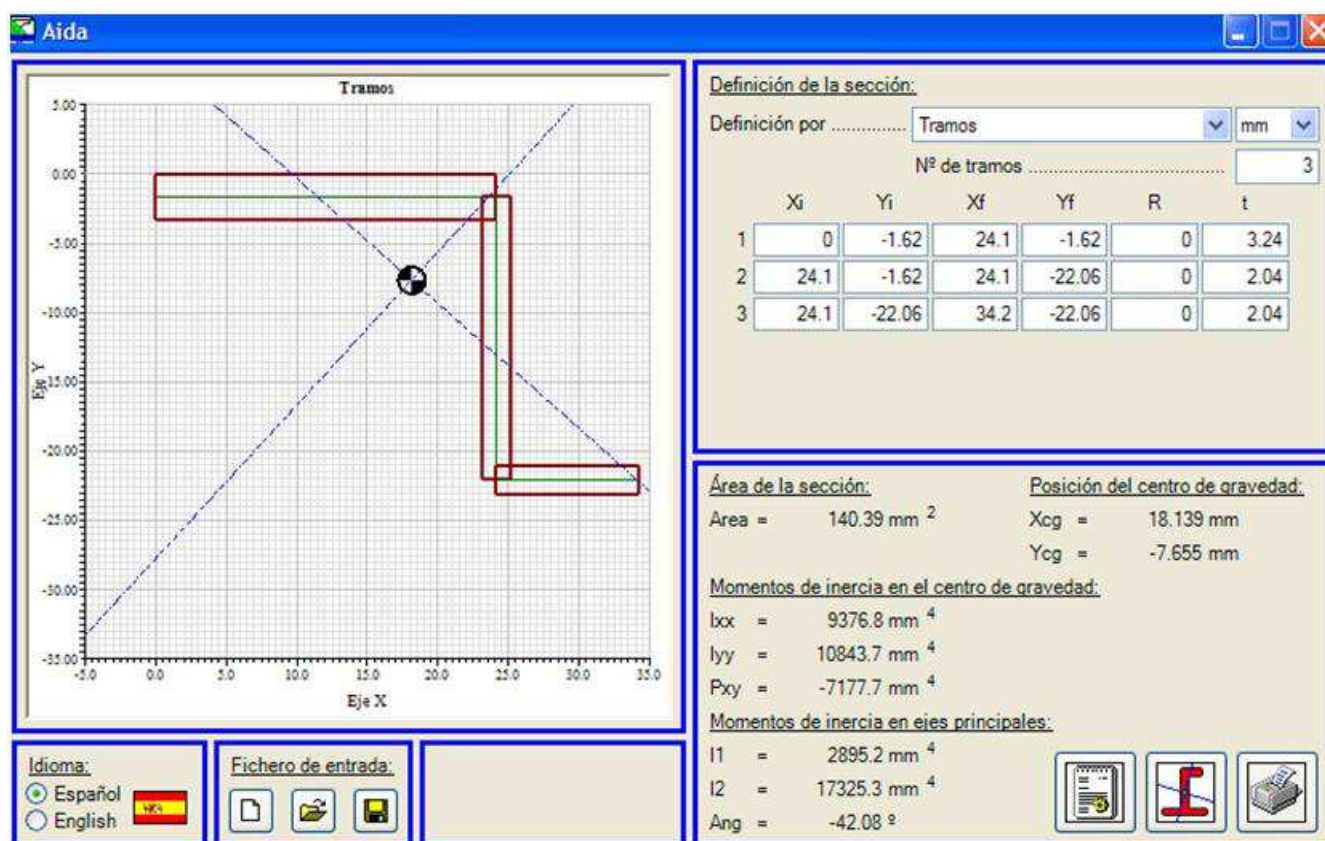


Figure 31. Example 9 Inertia properties

The upper and lower flanges are parallel to the direction of minor inertia, $I_p = 1$, and they have one end free and the other supported, $I_s = 1$. The web of the stringer is perpendicular to the direction of minor inertia, $I_p = 2$ and it has both ends supported, $I_s = 2$.

The input and output files corresponding to this example are shown on next pages:

8.9.1 Input file Example 9:

```

1 Program CRIPCO Example of utilization 9
  Stringer in "Z" of 480 mm length
  Section with three segments
  Vertical segment perpendicular direction minor inertia
2 Length
  480.0
3 Number of segments
  3
4 Segment data: Orient.  Apoyo  Anchura  Espesor  Ec  D11  D12  D22  D33  Ecu
  1 1   24.10   3.240  54932.1  148333.6  38753.2  155744.4  44147.6  10000.0
  2 2   18.92   1.800  62374.4  19899.7   5186.2   35157.4   6111.1  10000.0
  1 1   15.10   1.800  62374.4  19899.7   5186.2   35157.4   6111.1  10000.0
  
```

8.9.2 Output file Example 9

```
*****
*                                     *
*          <<< CRIPCO >>>          *
*                                     *
*****
* Calculation of the crippling allowable *
* value for composite stringers        *
*****
```

Title:
=====

Program CRIPCO Example of utilization 9
 Stringer in "Z" of 480 mm length
 Section with three segments
 Vertical segment perpendicular direction minor inertia

Input data:
=====

Length of the stiffener: 480.0

Number of stiffener segments: 3

Segment	Width	Thickness	Orient.	Supp.	Ec	D11	D12	D22	D33	Ecu
1	24.10	3.240	1	1	54932.1	148333.6	38753.2	155744.4	44147.6	10000.0
2	18.92	1.800	2	2	62374.4	19899.7	5186.2	35157.4	6111.1	10000.0
3	15.10	1.800	1	1	62374.4	19899.7	5186.2	35157.4	6111.1	10000.0

Allowable values for local buckling and crippling:
 =====

Segment	Local buckling	Crippling
1	-5267.7	-7199.7
2	-21541.0	-7199.7
3	-2895.0	-5961.7

```
*****
* Crippling microstrains for the stiffener: -6940.8 *
*****
```